

Time-Series Forecasting of Household Global Active Power Using Gradient Boosting Algorithms with Explainable AI Interpretation

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Abstract—Individual household electric power usage has been forecasted using a variety of time-series machine learning models. However, the performance of these models with the real-time data was poor. Along with that, some models' interpretation lacked genuineness. The real-world household data exhibit strong non-stationarity and irregular patterns of usage. Due to which, the classical time-series and deep learning models fail to capture them reliably. Also, the existing works focus either on reducing the error rate or on providing post-hoc explanations for the model's outcome, but they rarely address both aspects together in a unified framework for single-household forecasting. This creates a necessity for the models that perform better under real-time conditions and at the same time, exposing their internal decision logic in an interpretable manner. For the purpose of research, ensemble machine learning models such as CatBoost, XGBoost and LightGBM are taken into consideration for forecasting the electric power consumption in real-time. The proposed approach starts from a long-term, minute-level residential dataset. Data cleaning and hourly sampling is performed. By the means of feature engineering, lagged and rolling statistical features are created for the chosen gradient boosting regressors. The models are trained and their performance is measured with Root Mean Square Error (RMSE), Mean Absolute Error (MAE) and the coefficient of determination (R^2). Furthermore, explainable AI (XAI) uses SHAP to analyze the results in addition to displaying the best performing model's evaluation metrics. The results demonstrate that the top-performing ensemble machine learning model, LightGBM, provides accuracy and interpretability for household energy forecasting systems when paired with explainable AI. These findings highlight that a high-capacity ensemble model with principled explainability is a promising direction for trustworthy, data-driven demand-side management.

Keywords—Electric power, Time-series forecasting, Ensemble models, CatBoost, XGBoost, LightGBM, Explainable AI, SHAP.

I. INTRODUCTION

To understand how data changes over time, time-series forecasting is used, which helps track past patterns and predict future values. It is primarily used in data that is collected at regular intervals of time. It facilitates in identifying trends, seasonality and sudden changes. Due to which, it becomes useful for planning, prediction and decision making. It is applied to historical data that is consistent and reliable, when temporal patterns or trends exist and especially, when future behavior depends on the past observations. Time-series forecasting is widely used in weather, sales and sensor data.

As an ensemble learning approach, gradient boosting uses a combination of several decision trees to obtain more precise predictions. To improve prediction skills, this algorithmic method takes advantage of existing base models and builds upon their strengths. Gradient boosting is one of the best predictive modelling techniques because it can capture intricate patterns in data. Here, we use gradient descent to train each new model with the goal of minimizing the loss function, which can be anything from the previous model's mean squared error to its cross-entropy. The approach iteratively trains a weak new model to forecast the gradient of the loss function with respect to predictions. When a stopping requirement is satisfied, the ensemble (predictions from all models) is updated to include the new model's predictions.

Large-scale datasets with numerous independent features work well with CatBoost. Unlike competing gradient boosting techniques, CatBoost can automatically encode numerical and

categorical features. Symmetric Weighted Quantile Sketch (SWQS) approach in CatBoost helps in handling missing values, decreases overfitting and improves model performance. For optimal performance, speed and efficiency, XGBoost is used, an optimized gradient boosting optimization technique. Combining numerous weak models into a stronger one is the goal of this ensemble learning method. It can swiftly train models on massive datasets due to its integrated parallel processing. Users can optimize model parameters to solve specific problems with XGBoost's support for customizations. LightGBM is an ensemble learning system that prioritizes speed, scalability, and accuracy. Efficient decision trees are utilized, which minimize memory usage and optimize training time, allowing for efficient growth. When compared to competing frameworks, LightGBM's speed and accuracy are significantly enhanced by its pioneering features, such as Gradient-based One-Side Sampling (GOSS), histogram-based methods and leaf-wise tree growth.

A set of practices and methods known as explainable artificial intelligence (XAI) makes it possible for machine learning algorithms to generate results and outputs that humans can trust and comprehend. Companies that want to build trust when using AI can use XAI to their advantage. With the help of XAI, they can better know how an AI model operates and spot potential issues. One way to interpret a machine learning model's decision-making process is by using SHAP, an explainable AI framework. It shows the comparative importance of each input feature in influencing the outcome of the prediction. One can use it with any ML model. Assuring a constant and equitable distribution of feature priority is made easier with its support.

II. LITERATURE SURVEY

In [1], they came up with the RPLPM-GBX (Residential Power Load Prediction Machine Model-Gradient Boosting Regressor) model. It has a boosting regressor. The study found that the gradient-boosting regressor part of the Residential Power Load Prediction Machine model was really good at predicting energy use. It showed a MAE value of 0.0222, MSE value of 0.0012 and RMSE value of 0.0350.

[2] demonstrated the effectiveness of the LightGBM model in predicting household energy consumption using high-resolution temporal data, such as hourly energy usage throughout the year. The proposed approach achieved an RMSE of 0.501, an MAE of 0.420, and an MSE of 0.251. The study further highlighted that the feature `sub_metering_3`, representing the electricity consumption of water heaters and air conditioners, had a substantial influence on the model's predictions. This finding underscores the critical role of sub-metering information in enhancing the accuracy of energy consumption forecasting systems.

In regards to [3], they conducted a comprehensive comparison of seven machine learning and deep learning models (LR, RFR, LGBMR, XGBR, CBR, LSTM, BiLSTM) for household energy consumption prediction. They used information from Bangalore, India that they collected from April to August in 2021. They collected this information every 20 minutes. The LightGBM Regressor model was really good at guessing the amount of energy used. The RMSE value was 0.0635, the MAE value was 0.0318 and the R^2 value was 0.7538. It is found electrical current and the lagged energy consumption were the most important features that affected the predictions.

In [4], they performed a relative analysis between five decision tree-based learning models. They are Random Forest, Gradient Boosting Machine, XGBoost, LightGBM and CatBoost. They worked on a couple of datasets. One set of data is from their University Residential Complex. The other set of data from the AEP (Application Energy Prediction) dataset. The University Residential Complex dataset showed that GBM did hourly prediction with an average error of 5.86%. On the other hand, the AEP dataset showed that CatBoost was really good with hourly prediction with an average error of 48.23%. They found that the temperature-humidity index and the wind chill temperature were features that dominated in the hourly prediction of electric power consumption.

In [5], the authors conducted a comprehensive review of explainable artificial intelligence (XAI) techniques that are used in energy and power systems. The authors highlighted that XAI enables feature importance analysis, model debugging and user trust-building. On the other hand, there were certain key challenges that were observed in regards to the trade-off between predictive accuracy and interpretability and lack of standardized explanation evaluation metrics.

In [6], they presented a household power consumption prediction method using selective ensemble learning, K-means clustering and a novel Filtering Iterative Optimization Ensemble Strategy (FIOES) to monthly consumption data. The results show that FIOES achieves prediction error in the range of 18-23% with an average error of 7.35%, which clearly outperforms single-base learner models. However, explainable AI techniques or frameworks are not used to understand the model's interpretability. Due to which, the model failed to create authenticity and transparency among the users involved in this domain.

III. PROPOSED METHOD

The proposed research work aims to forecast the Global Active Power for individual households and to perform a comparative analysis of three gradient boosting regression models. They are: CatBoost, XGBoost and LightGBM. In order to know their interpretation, an explainable AI framework, SHAP is used.

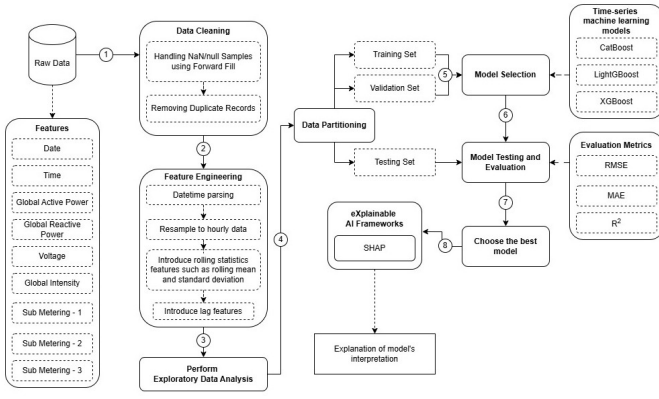


Fig. 1. Block Diagram of the Proposed System

Fig.1. explains the complete architecture of the proposed system. The overall pipeline consists of five stages. Data acquisition, data cleaning for removing NaN, Null or duplicate samples, exploratory data analysis for determining the pattern of each feature with respect to time and to know if there is any dependency found between any of the features, feature engineering with respect to the model, dataset partitioning into training, validation and testing sets, individual model training and evaluation and interpretation of the models' outcomes using SHAP. By combining rigorous preprocessing of a real-world high-frequency household energy dataset, tailored time-series feature engineering, comparative training of advanced gradient boosting models and explainable AI-based interpretation using SHAP, the proposed method addresses both the accuracy and interpretability requirements of individual household Global Active Power forecasting.

PSEUDOCODE OF THE PROPOSED METHOD

Input: Minute-level individual household electric power consumption dataset, D_{raw} .

Output: Best predictive model, M_{best} and feature importance, FI.

Begin

- 1) Load dataset, D_{raw} .
- 2) Create Datetime from Date and Time features.
- 3) Sort D_{raw} by Datetime.
- 4) Handle missing values using forward fill.
- 5) Remove duplicate Datetime records.
- 6) Aggregate data to hourly level and store the result as D_{hourly} .
- 7) Set target variable, $y \leftarrow D_{hourly}, global_active_power$.
- 8) Construct feature matrix, X using:
 - lag features of y .
 - rolling statistics of y .
 - time-based features from Datetime.
- 9) Split (X, y) sequentially into training set, validation set and testing set.
- 10) Define candidate models:
 - Models $\leftarrow$ XGBoost, CatBoost, LightGBM.

11) For each model, M in Models:

- Tune hyperparameters using training and validation sets.
- Train M on combined training and validation data.
- Evaluate M on test data.
- Store performance metrics.

12) Select best model, $M_{best} \leftarrow$ model with minimum RMSE, MAE or maximum, R^2 .

13) Compute SHAP values for M_{best} and derive global feature importance, FI

14) Output performance metrics and FI of M_{best}
End

IV. METHODOLOGY

A. Data Collection: The first stage is to gather the raw data collection known as the "Individual Household Electric Power Consumption." This dataset contains around 2,075,259 minute-level records of a single household in Sceaux, France, spanning the years 2006–2010. There are nine features in the raw dataset. Among these are the following: date, time, global active power, global reactive power, voltage, global intensity, and three sub-metering headings, namely sub_metering_1, sub_metering_2 and sub_metering_3, which correspond to the kitchen, laundry room, and a combined water heater and air-conditioned circuit, respectively. The date and time attributes are merged together into a single datetime feature and converted to a standard datetime format suitable for performing time-series operations. Missing samples are handled using the forward-fill method for ensuring the temporal consistency without discarding large portions of data. Redundant entries are checked with respect to the datetime index and removed thereby. To preserve daily and seasonal consumption patterns, the minute-by-minute series is resampled to hourly data by aggregating the values of each individual feature found in the dataset.

B. Feature Engineering: Using the hourly-resampled data, additional temporal features are derived from the datetime index. Rolling mean and rolling standard deviation are the statistical features that are computed over selected window sizes for the global_active_power to capture local trends and short-term variability. To learn temporal dependencies and autoregressive structure in household consumption, lag features of the target variable are introduced. This results in the formation of a multivariate supervised learning dataset that is suitable for gradient boosting regressors.

C. Dataset Partitioning, Training, Validation and Evaluation of the chosen Ensemble Models: In the third stage, the processed dataset is partitioned into training, validation and testing sets. For model fitting and hyperparameter tuning, the training and validation sets are used. To ensure a fair comparison, the same feature set is used by the three ensemble methods, XGBoost, CatBoost and LightGBM, for predicting the target variable, that is,

Global Active Power. The models' performance is evaluated on the testing set using the standard regression metrics: Root Mean Square Error (RMSE), Mean Absolute Error (MAE) and coefficient of determination (R^2). This allows a direct comparative analysis of the three gradient boosting models under identical data and evaluation conditions.

- **Root Mean Squared Error (RMSE):** The square root of the mean of the squared difference between the actual and predicted values is calculated using RMSE as follows:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

- **Mean Absolute Error (MAE):** The absolute difference between the actual and expected values is quantified by the MAE. It is computed as follows:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- **Coefficient of Determination (R^2):** The amount of the dependent variable's variance that each of the independent variables account for is calculated using R^2 . Its purpose is to quantify the goodness of fit of a regression model. It is calculated as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

D. Explainable AI Interpretation: The final stage integrates explainable AI techniques to interpret the predictions produced by a model that shows relatively best performance. SHapley Additive exPlanations (SHAP) are applied to compute global feature importance values. It facilitates in revealing which features strongly influence the model's output across the entire dataset. SHAP forms an explainability layer on top of the chosen gradient boosting model enabling transparent analysis of the forecasted output.

V. RESULTS AND DISCUSSIONS

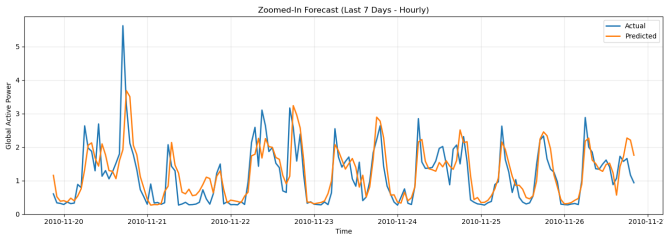


Fig. 2. Zoomed View of actual versus CatBoost-predicted Global Active Power for the last seven days of the test set.

From Fig.2. we find that the CatBoost model has performed worse in this dataset. Due to which, the model has resulted

with RMSE and MAE values of 0.4420 and 0.3064, respectively. These error magnitude values are relatively high when compared to the other chosen ensemble time-series forecasting machine learning models. Its coefficient of determination, R^2 , holds a value of 0.6067 which clearly shows that the model is a weaker fit to the true consumption pattern.

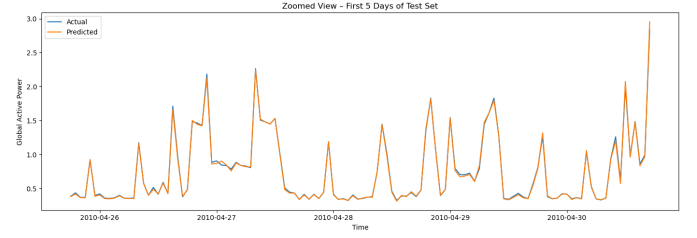


Fig. 3. Zoomed View of actual versus XGBoost-predicted Global Active Power for the first five days of the test set.

It is clear from Fig. 3. that among the ensemble models that were chosen for this research, XGBoost showed a good result in the prediction process. It has a RMSE value of 0.0252 and MAE value of 0.0159. The coefficient of determination, R^2 , turns out to be 0.9987. This clearly shows that the model is almost-perfect to the true consumption pattern of the real-time individual household electric power consumption data.

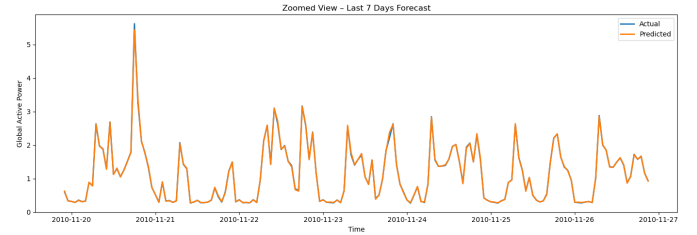


Fig. 4. Zoomed View of actual versus LightGBM-predicted Global Active Power for the last seven days of the test set.

Fig. 4. shows that the LightGBM shows the best performance among the models that were taken into consideration. It holds the lowest RMSE and MAE of 0.0218 and 0.0135, respectively. It holds the highest value for the coefficient of determination, R^2 , that is about 0.9990.

TABLE I
COMPARATIVE EVALUATION OF GRADIENT BOOSTING MODELS

Models	RMSE	MAE	R^2
LightGBM	0.0218	0.0135	0.9990
XGBoost	0.0252	0.0159	0.9987
CatBoost	0.4420	0.3064	0.6067

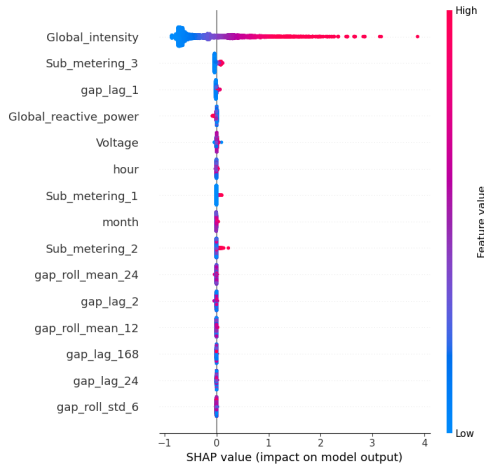


Fig. 5. SHAP summary plot illustrating the global feature importance and impact of input variables on the LightGBM model predictions for Global Active Power.

Fig. 5. shows the SHAP summary plot that illustrates the global feature importance of the best performing model, LightGBM, for household electric power consumption forecasting. The feature globalintensity emerged as the major predictor, while secondary features like sub_metering_3 and gap_lag_1 contributed moderately.

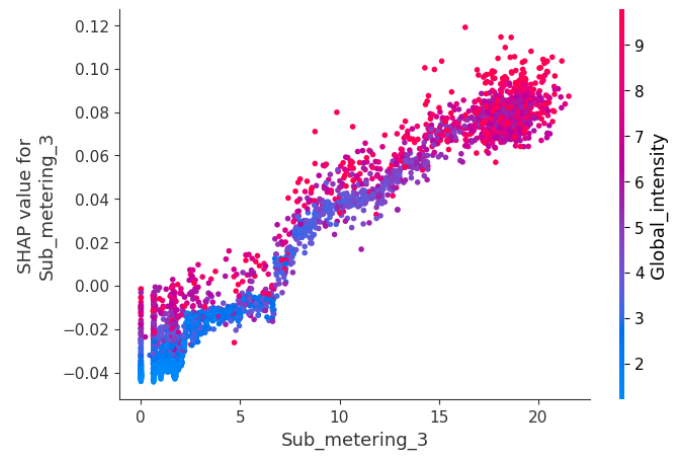


Fig. 7. SHAP dependence plot showing the contribution of the sub_metering_3 feature to the LightGBM model output

From Fig. 7, it is clear from the SHAP dependence plot that there is a strong linear positive linear relationship between sub_metering_3 and household electric power predictions. Low sub_metering_3 magnitudes indicate a suppressed consumption forecast. However, there is a strong positive influence on predicted power consumption with increase in value of sub_metering_3.

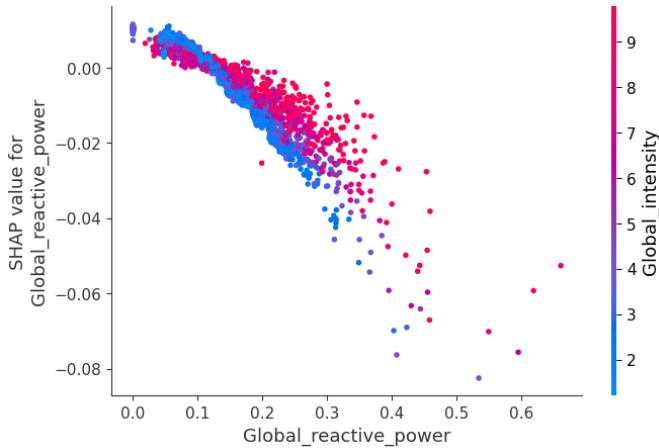


Fig. 6. SHAP dependence plot showing the contribution of the global_reactive_power feature to the LightGBM model output.

Fig. 6. shows a non-linear and inverse relationship between the global_reactive_power and global_active_power by the means of SHAP dependence plot. It reveals that low global_reactive_power has less impact on consumption forecasts. However, global_reactive_power with higher magnitude has decreased the overall consumption of electric power in the household.

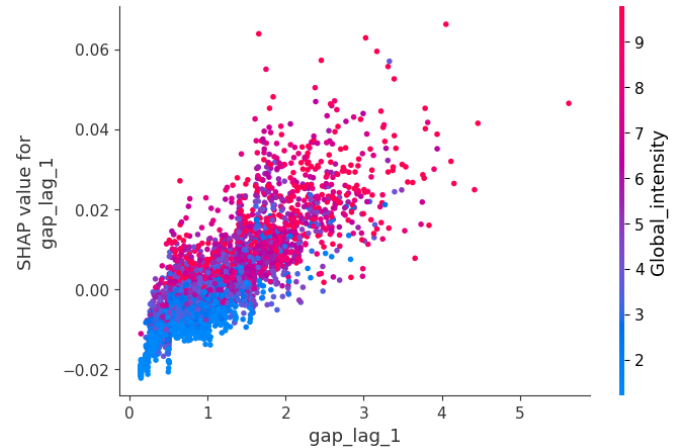


Fig. 8. SHAP dependence plot showing the contribution of the gap_lag_1 feature to the LightGBM model output

The SHAP dependence plot in Fig. 8. shows the non-linear relationship between gap_lag_1 and the model predictions when the values of gap_lag_1 are low. It is noticed that higher the value of gap_lag_1, there is an establishment of strong positive correlation with the predicted values of the electric power consumption.

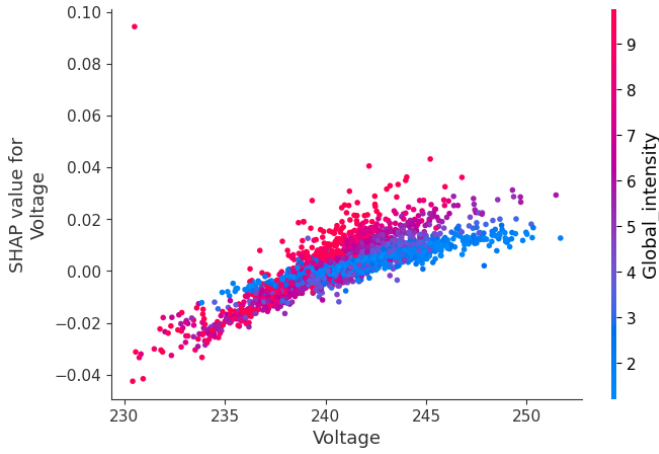


Fig. 9. SHAP dependence plot showing the contribution of the voltage feature to the LightGBM model output

Fig. 9. has a SHAP dependence plot that illustrates the weak and positive linear relationship between voltage and predictions of the model. It is observed that there are suppressed consumption forecasts when the voltage values are low. Also, it is clearly seen that higher voltage supports increased predicted power consumption.

TABLE II
GLOBAL FEATURE IMPORTANCE OF THE LIGHTGBM BASED ON MEAN ABSOLUTE SHAP VALUES

Rank	Feature	Mean absolute SHAP value
1	global_intensity	0.562263
2	sub_metering_3	0.041239
3	gap_lag_1	0.010053
4	global_reactive_power	0.006441
5	voltage	0.005782

Since LightGBM has shown the best result among the chosen gradient boosting models for real-time forecasting of the individual household electric power consumption, explainability AI technique, SHAP, is used in order to understand the model's interpretation. SHAP Explainability AI framework reveals that global_intensity feature contributed the most for the prediction with an absolute mean SHAP value of 0.562263. Followed by, sub_metering_3 feature has contributed in the household electric power consumption prediction with a mean absolute SHAP value of 0.041239.

VI. CONCLUSION

The research demonstrated that gradient boosting ensemble models, when combined with explainable AI, can accurately and transparently predict and forecast individual household global active power. The model that performed the best is the LightGBM. Closely followed by is the XGBoost. CatBoost was not able to generalize for the scenario that we are concerned about. The process of acquisition and cleaning of data, resampling and engineering the features of the dataset

allowed the models to capture and understand the complex underlying patterns effectively. To maintain the transparency and understand the model's interpretation, the explainability framework, SHAP, revealed global_intensity as the major dominating variable in the prediction of individual household electric power consumption. This modular and scalable approach can be extended to multi-household settings and enriched with other necessary features to further support data-driven residential energy optimization.

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